

IBM Resiliency Summit

z/OS High Availability: Achieving high resilience, designed to provide continuous operations for mission-critical workloads

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Z/OS Resilience Development Team



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Agenda

1. Resiliency Basics – where do you need to be?
2. Parallel Sysplex a must for High Availability
3. Highlights of z/OS Resiliency features
4. z/OS Workload Impacting events that we see
5. Commonly Not Followed – Best Practices
6. What is z/OS doing about these issues?
7. Less advertised “recent” z/OS resiliency improvements
8. Reference material - *IBM z Content Solutions Center*

Determine your Workload Recovery objectives!

The resilience architecture must meet the workload’s SLA (Service Level Agreements) tolerations for data loss and lack of service for both planned or unplanned outage:

- SRO (Service Restoration Objective) : the **target time** for how quickly a service will be restored for either planned or unplanned outages
- RTO (Recovery Time Objective) : **maximum acceptable time** from the loss of service until it is restored
- RPO (Recovery Point Object) : **maximum amount of data loss** measured in time since the last backup or recovery point

(High) availability		Disaster recovery
Triggering Event	Workload component failure within a datacenter	Event that results in a full Datacenter outage
Solution	Restore full service within the datacenter.	Re-establish critical services at an alternate site.
Typical Recovery Objectives	• SRO = seconds or minutes • RPO = 0 SRO “clock” starts when incident occurs	• RTO = hours or less* • RPO > 0 or = 0**, measured in minutes RTO “clock” starts when disaster is declared

*GDPS Active/Active Continuous Availability provides near-continuous availability across geographically dispersed sites
** GDPS Continuous Availability with Zero Data Loss (ZDL)

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All things to consider

Vision

Strategic Alignment & Foresight

- Aligns with Goals and Regulatory Requirements
- Defines SRO, RTO, RPO, and SLAs
- Supports **resilience scoring** and **continuous improvement metrics**

Proactive Management

Operational Readiness

- Automated Monitoring
- Disciplined Maintenance Strategy
- Predicative Failure Monitoring
- AI-driven Testing, Capacity Planning, and Change Management
- Integrated with intelligent remediation workflows for streamlined issue resolution

Cohesiveness

System Interoperability

- Cross-Platform Integration
- Application Dependency Mapping
- Data Interdependence Insights
- Supports Hybrid Cloud Resilience and API-driven orchestration

Technology

Infrastructure Resilience

- Fault-Tolerant HA Architecture “eight 9s” with Concurrent Repair/Upgrade and no SPOF
- Data & application sharing across multiple systems
- Ability to swap between machines and datacenters to provide rapid failover and proactive avoidance

Perseverance

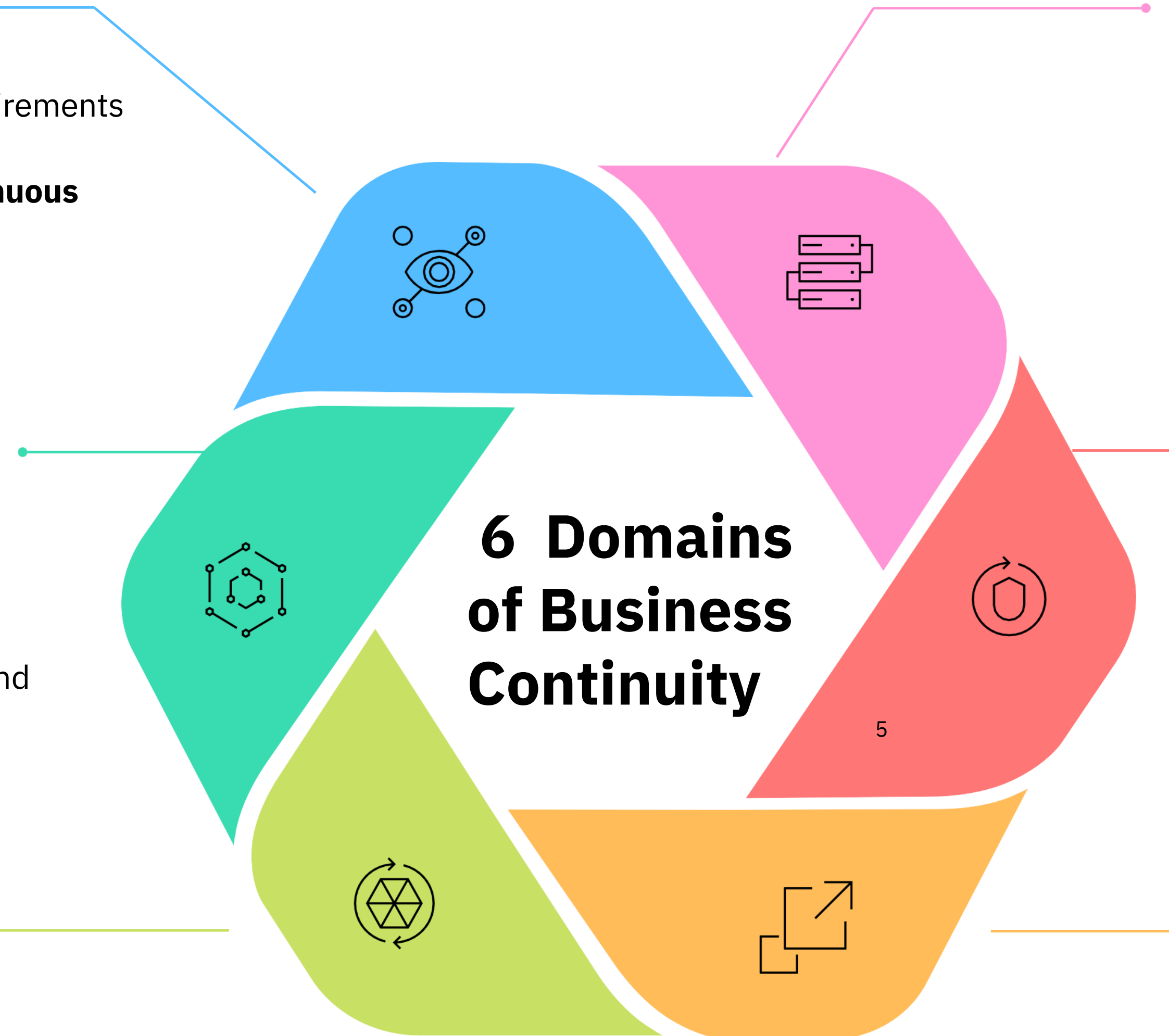
Security & Continuity

- Pervasive Encryption, Redundancy, and Immutable Backups
- [Dynamic Routing, Workload Balancing, and Autonomic Operations](#)
- Disaster Recovery Automation and Data Replication
- Protection and recovery from cyberattacks / ransomware

Performance

Scalability & Efficiency

- [Autonomic Capacity Provisioning \(Scale Up/Down\)](#)
- Latency Optimization and I/O Performance Management
- AI-accelerated Workload Insights
- Scalable, consumption-based HW/SW



z/OS Parallel Sysplex “Redundancy/Cloning” for High Availability

With HyperSwap, enables application availability of 99.99999%*

Removes Single Point of Failure of:

- Servers
- LPARs
- Subsystems

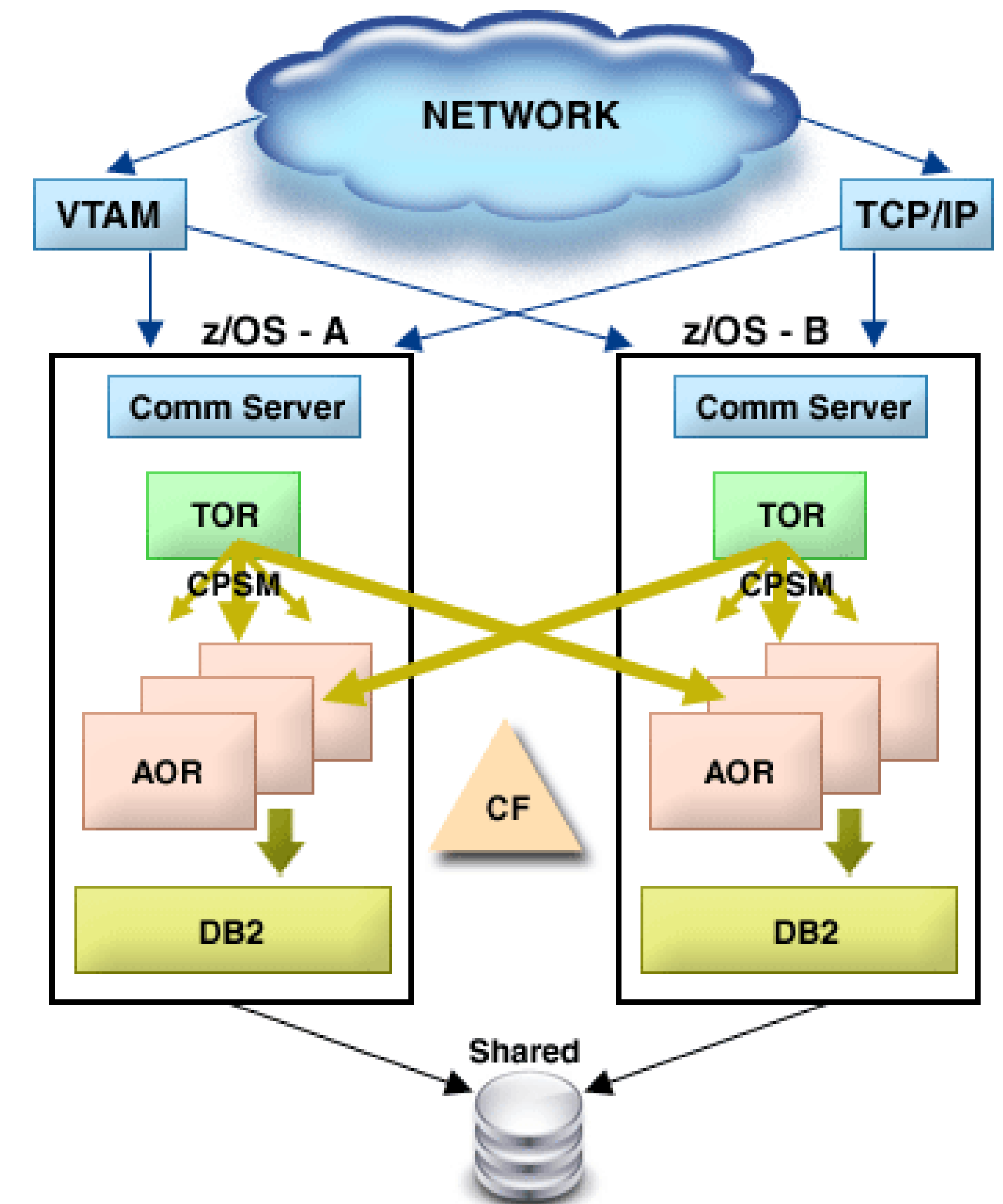
Can be leveraged for planned or unplanned outages

Presents a single system image to

End user,
Operations,
System Programmer

Uses Dynamic Workload Balancing to maximize system performance

Well suited to assist with workloads managed by key subsystems such as CICS, IMS and Db2



Highlights of IBM Z Stack Innovations in Resiliency

Faster recovery when failures occur

- ✓ CICSplex, DB2, VSAM RLS, IMS: Datasharing, MQ Queue Sharing, ...
- ✓ z/OS DVIPA, Sysplex Dynamic Workload Routing (Sysplex Distributor...)
- ✓ z/OS zCX Sysplex Distributor, Red Hat® OpenShift® cluster, Elasticsearch (ELK)
- ✓ z/OS VSAM RLS for Catalog, Shared zFS, PDSESHARING(EXTENDED) i.e. write
- ✓ System Recovery Boost (SRB) (IPL, Middle Ware stop/restart support, Dump, Hyperswap, dynamic I/O changes)
- ✓ Flexible Capacity, Capacity BackUp (CBU)
- ✓ z/OS AutoIPL, Automatic Restart Manager (ARM)
- ✓ z/OS Sysplex Failure Management (SFM), CFLCRMGMT XCF Function
- ✓ z/OS BCPII System Status Detection Partitioning Protocol
- ✓ CF: External CF, CF System/User/Asynch Managed Duplexing
- ✓ zHyperLink (VSAM Verify), DFHSM Dataset (z/OS & zFS File) Backup
- ✓ IBM Z Backup Resiliency, DB2 advance recovery, IMS FDBR
- ✓ z/OS IOS CU Recovery
- ✓ Z16 CF Improved IFCC Handling
- ✓ IBM Z Cyber Vault

Failure Avoidance

- ✓ Test: Accelerator for Z (IBM TST), Stress: IBM ZBURST
- ✓ Operations: IBM Concert For Z
- ✓ z/OS Blocked Workload Support (BLWLINTHD, BLWLTRPCT)
- ✓ z/OS RSM address space Dedicated Memory
- ✓ GRS ENQ Contention Analysis: RMF, Intellimagic, etc.
- ✓ z/OS XCF Message Isolation (MSGISO), CF Monopolization (CFMONOPAVOID)
- ✓ z/OS Syslog Message Flood Automation
- ✓ SMF Flood Automation
- ✓ Capacity Provisioning
- ✓ XCFCNFXX- LOGR Services Monitor
- ✓ Auto-WTOR Response
- ✓ Cyber Resiliency,
- ✓ IBM Thread detection for z/OS (TDz)
- ✓ TFP-HW
- ✓ HMC LPAR Error Protection
- ✓ Z17 HMC Dual Control

Problem Determination and Data Capture

- ✓ Run Time Diagnostics (RTD)
- ✓ Dump cmd Parmlib members (IEADMCxx)
- ✓ AutoIPL w/Standalone Dump
- ✓ z/OSMF Incident Log
- ✓ SMF Logstreams (Performance & Scalability)
- ✓ SMF 98s, IBM Concert: zWorkload Information Correlator (zWIC)
- ✓ SMF 98s, IBM Concert: zWorkload Information Navigator (zWIN)
- ✓ New SMF Records & Fields

React faster to workload fluctuations

- ✓ zHyperLink, zHyperwrite, DFSMS Striping
- ✓ FICON 32S
- ✓ Capacity Provisioning Manager (CPM)
- ✓ Tailored Fit (TPF-HW) for HW, On/Off Capacity On Demand (OOCOD)
- ✓ Hiperdispatch
- ✓ More Memory (Larger Buffer Pools, GBPs...)
- ✓ Z17 Performance Improvements

Improved workload scaling

- ✓ Hiperdispatch
- ✓ zHyperLink, zHyperwrite, Read from Secondary, zEDC
- ✓ JVMs pause-less garbage collection
- ✓ FICON 32S, z17 CL6 Throughput improvements
- ✓ Large Page Support, VSAM RLS for Catalog
- ✓ CICSplex, DB2, IMS Datasharing, MQ Queue Sharing
- ✓ XCF XTCSIZE, zOS V1R10 VSM algorithm improvements, V1R8 RSM
- ✓ Sysplex Distributor, VSAM RLS 2 Lock Structures
- ✓ z16 CF Adapter Performance Improvements
- ✓ Z16 CFCC Code Scalability Enhancements
- ✓ DFSMS Extended Addressability, DFSMS Striping

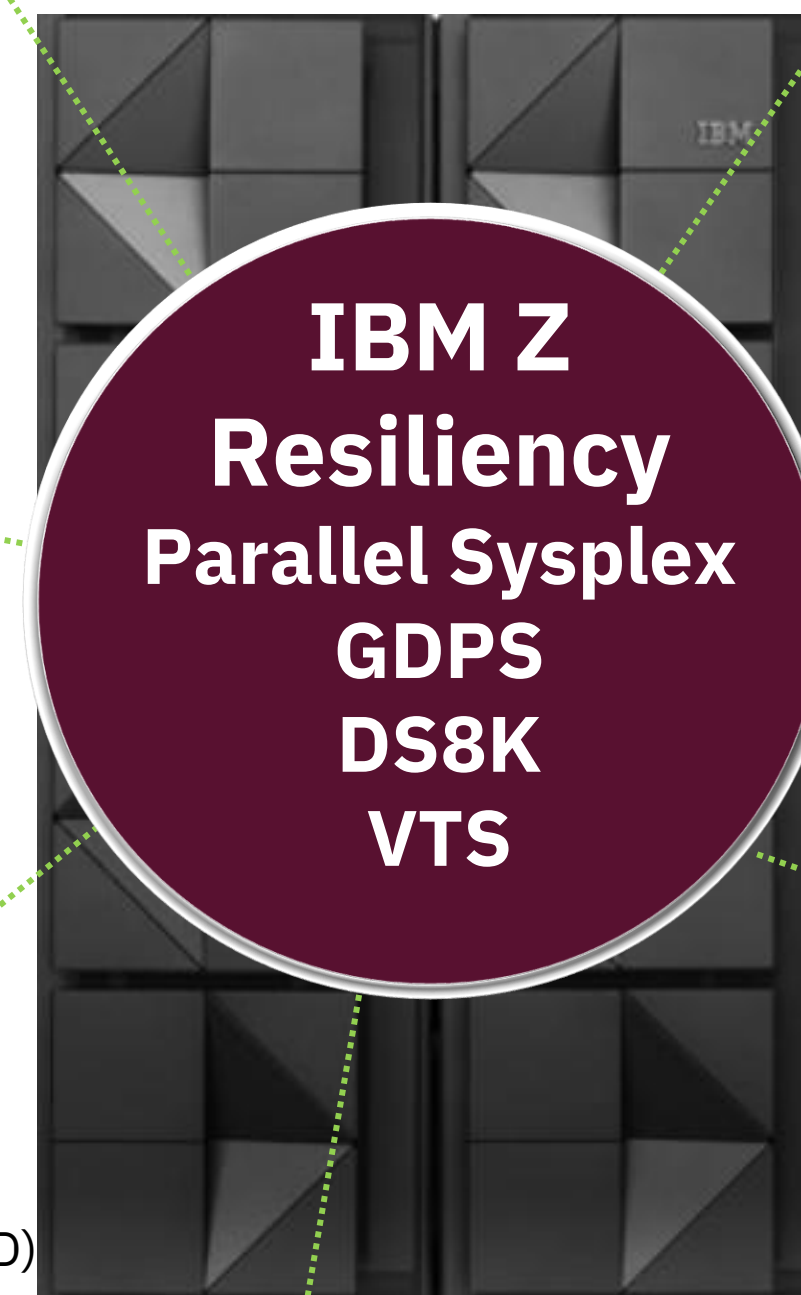
Concurrent maintenance

- ✓ Software coexistence and rolling IPLs
- ✓ z/OS concurrent maintenance, Fixcats
- ✓ Dynamic I/O Reconfiguration
- ✓ HyperSwap

Preventative / Real-Time Insight

- ✓ z/OS Health Checker
- ✓ Predictive Failure Analysis (PFA)
- ✓ IBM Z Anomaly Analytics (IZAA now part of IBM Concert for Z)
- ✓ IBM Concert For Z
- ✓ IBM Z Operational Log and Data Analytics
- ✓ Heartbeat Monitoring – Omegamon, RMFMON, Intellimagic, OpenTelemetry

And many more...



Insights from z/OS and GDPS incidents over the past few years

- **“User errors” contributed to approximately 50% of the incidents found in production and test environments**
 - **Change control:**
 - Insufficient test of changes, failure scenarios, rate increases etc. : z/OS, GDPS, new applications, middleware, ...
 - Certificate management: mostly expiration issues but also general rollout of new ones not done properly
 - Device management issues
 - **Service not applied**
 - 80% of these were HIPER with an available fix : 75% fix available < 6 months with 25% > 6 months
 - Follow strategy outlined in “z/OS Preventive Maintenance Strategy to Maintain System Availability”
 - Enhanced HOLDDATA, monitoring HIPERs and applying other service (RSUs)

z/OS Workload impacting incident insights (Cont)

- **“User errors” continued**, many fall into various areas with a small number of incidents:
 - **Network configuration errors/complexity**
 - **Not using a preventative resilient technology**
 - Datasharing, GRS STAR, Metro Mirror, 64-bit LDAP, ...
 - **Insufficient capacity planning:**
 - Workload increasing but not resources: CPU, Real Memory, Aux storage, Virtual Storage Constraint, ...
 - **z/OS Health Checker not used properly**
 - Check exceptions not handled properly or never considered
 - **Incorrect Catalog Sharing Serialization setup**
 - **Power outages : back-up not setup or configured properly**

z/OS Workload impacting incident insights (Cont)

- **“User errors” continued**, many fall into various areas with a small number of incidents:
 - **“Best practice config settings” not followed:**
 - Bad Parmlib change
 - USS Sysplex Root mounted RW rather than RWS or RO
 - Using HW Reserve rather than global ENQ
 - Hyperswap not being used
 - **Operational Procedure Errors:**
 - Full JES2 spool not handled properly
 - Ran out of ASIDs: Health check ignored, poorly planned “therapeutic” IPLs, REUSASID=YES not specified for frequently restarted STCs.
 - ...

z/OS Workload impacting incident insights (Cont)

- **Serialization issues are approx. 6% of problems:**
 - **Monitor ENQ and other contention for growth or abnormalities especially in test**
 - RMF, Omegamon, SMF 98 (zWIC/zWIN), ...
 - **Ensure operational planning includes “good” contention analysis**
 - Contention analysis should be across the stack i.e. middleware
 - For z/OS Consider using RTD (Runtime Diagnostics), and other tools to identify and remove blockers
 - RTD covers GRS ENQ and latch services, as well as USS specific monitoring for their latches
 - D SMS, SMSVSAM,DIAG(CONTENTION) cover RLS and other SMSVSAM related latches
- **Other monitors RMF, middleware monitors, Omegamon, etc.**

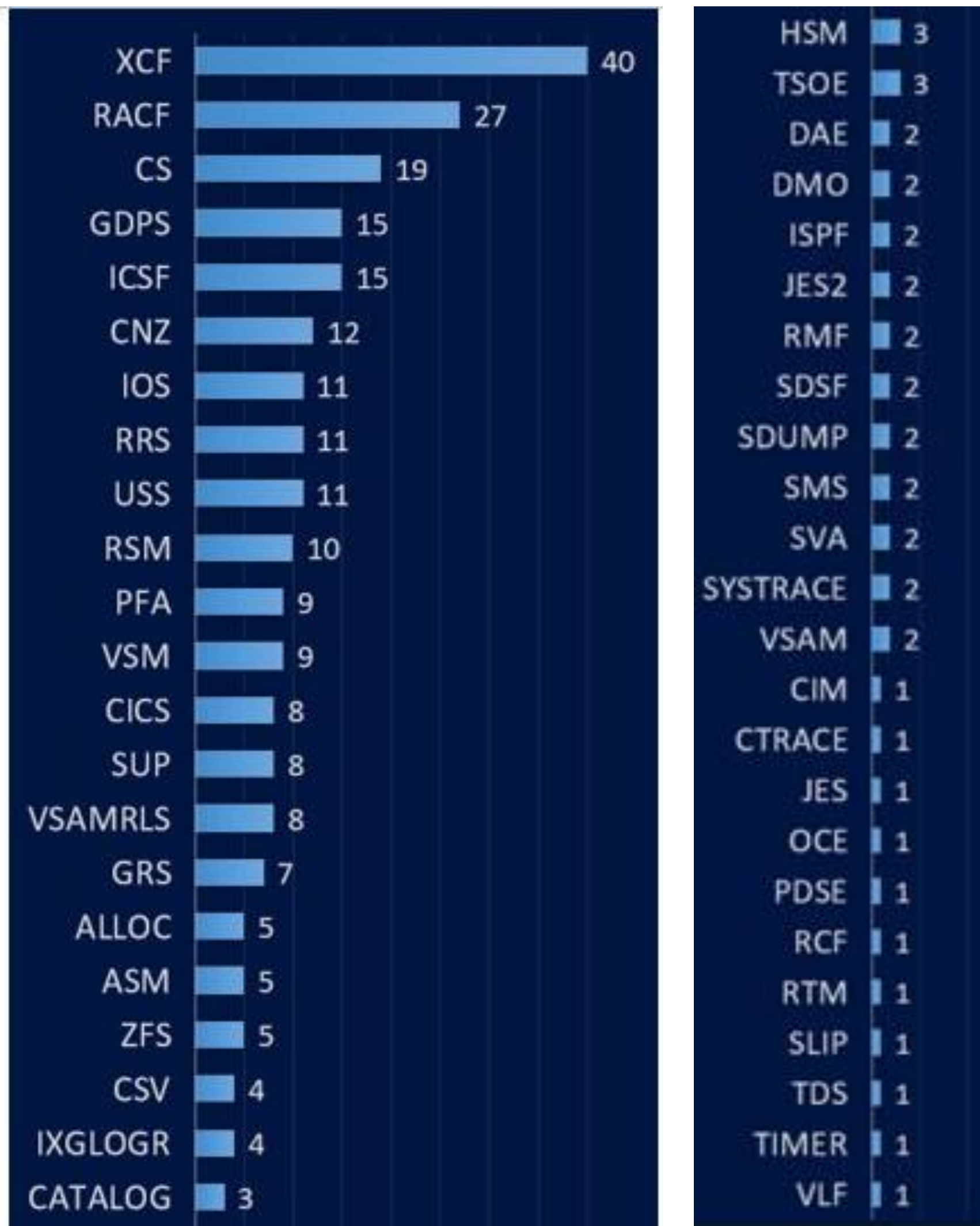
Commonly not followed “Best Practices” noted by IBM WSC

Common issues/best practices are not implemented frequency from all studies

- z/OS Health checker exceptions not integrated into monitoring process. **HC severities are lowered.**
 - Track new service stream enabled Health Checks via the IBM.Function.Health fixcat
 - Monitor and automate for Health Check alerts, note: PFA triggers are health check alerts
 - Check your policy to make sure you re-activated temporarily deactivated checks
 - Do not ignore LOW importance exceptions
- XCF functions : CFLCRMGMT and several others not used.
 - See White Paper: Best Practices for IBM z/OS in a Parallel Sysplex
- SRB middleware stop/restart not used (WLM policy update), SRB dump support not implemented
- Storage Support for DASD Protection (SSDP) not implemented
- ARM (Automatic Restart Manager) not implement
- Logger IXGCNFX Logr monitoring not active
- Ctrace Traces are left active; there is a health check for that!
- VSMBESTFITCSA(YES) not coded
- USEV1R19Rules(YES) coded, but we want NO! Make sure you test as incompatible!
- SWBSTORAGE(ATB) not used: 64 bit SWB (Scheduler Work Block)
- VTOC_USE_ZHPF not used
- Long distance CF Adapter used for short distance
- CF Performance not monitored, reported, tracked
- Memory ‘minimum available frame queue’ not monitored

Common issues: z/OS Health Checker

Of Health Checks Group By Owner



Close to 300 Base Health Checks exist to help improve your resiliency.

How to Print Health Checks Information in Batch Mode

```
SYS1.SAMPLIB(Hzsprint)
```

```
//HZSPRINT EXEC PGM=HZSPRNT,TIME=1440,REGION=0M,PARMDD=SYSIN  
//SYSIN DD *  
CHECK(*,*) ,EXCEPTIONS  
//SYSOUT DD SYSOUT=A,DCB=(LRECL=256)
```

CHECK(*,*) Shows all HCs

CHECK(*,*) , EXCEPTIONS Shows HCs with exceptions

Common Issues : Questions to ask

1. Are you tracking IBM.Function.Health Checker fixcat ?
2. Are you monitoring HC alerts ? Do you have automation for HC alerts ?
3. When did you last check your policy to make sure you activated back HCs that were temporarily deactivated before?
4. Did you deactivate or decrease the priority of critical HC – I know let me check that later?
5. What is the default setting for new HCs according to your policy ?

What are we doing about these issues?

- **z/OS resiliency team:**

- Reviews workload impacting problems on a regular basis to drive enhancements
- Drives and approves various resiliency initiatives

- **Simplification:**

- Major Initiative: Project Polaris and simplification efforts – next chart!
- In addition to Polaris: Some noted Best practices improvements
 - 3.1, 3.2 WLM Policy Advisor – checks your policy for common problems
 - 3.2 USS_ENVIRONMENT check for USS Best Practices
 - Note ZOSMIG32_* checks indicating depreciated parms and default changes

What are we doing about these issues? (cont)

- **Problem avoidance improvements:**

- Improved customer testing capabilities
- Resiliency Posture understanding

- **Operational improvements**

- IBM Concert , Open Telemetry, ...
- Improve comprehensive contention analysis across the stack
- Root cause analysis

- **New “Getting Started with Business Continuity on IBM Z” redbook:**

<https://www.redbooks.ibm.com/redpieces/abstracts/sg248446.html>

- Covers all aspects to consider and related technologies

IBM Project Polaris

The Next Generation z/OS Experience

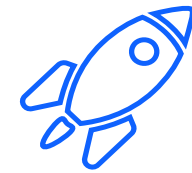


A **simplified z/OS user experience** based on the needs of early tenure mainframe professionals.

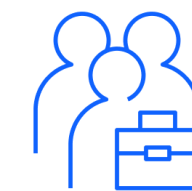
By integrating AI-driven insights and enabling the use of industry-aligned tooling, the new z/OS user experience is designed to make z/OS more intuitive and accessible.

For more information on IBM Project Polaris and the technical preview program, sign up here: ibm.biz/PolarisSUProgram or email zos@ibm.com

IBM
Project
Polaris Key
Value



Faster time to proficiency – Reduces the z/OS learning curve from years to months, accelerating productivity and adoption of critical Z technologies



Easier to use for new users– Early-tenure workforce shifts and skills transition increases the demand for more intuitive, guided experiences



Modern, industry-aligned experience – Aligns with common development practices and tools to make the platform more accessible

IBM Project Polaris

The Next Generation z/OS Experience

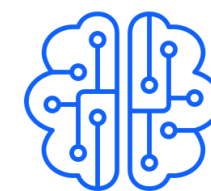


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For more information on IBM Project Polaris and the technical preview program, sign up here:

ibm.biz/PolarisSUProgram or email zos@ibm.com



- **Configuration:**

- **as-code model:** Centralized configuration registry simplifies complex PARMLIB syntax, bringing source code management rigor to z/OS configuration. Enables GitOps including version control and automation pipelines for consistency, auditability, and repeatability

- **Simplify:** remove unused/needed options, change defaults to best practices, more health checks when not following best practice

- **AI at the core** – Agentic AI and watsonx Assistant for Z integration enables contextual expert-level guidance, broaden expertise, boost efficiency, and task completion with confidence
- **API-first foundation** – As the primary interface for interacting with z/OS, APIS are expanded for simplified access to system information while enabling automation, integration, and infrastructure-as-code practices
- **Intent-driven interactions** – Dynamic, context-aware interfaces aligned to tasks rather than screens. Enables configuration-driven workflows, CLI usage, AI-infused UIs, and integrated tooling that reduce learning barriers

Less advertised z/OS “resiliency” improvements

- **VSCR (Virtual Storage Constraint Relief) continues**

- 3.2 DIAGxx ASCBV31=YES moves selective SQA ASCBs to ESQA
- 3.2 DIAGxx CBLOC VIRTUAL31:
 - IHAXTLST moves module extent list for LPA (CSA->ECSA) and private modules (private->extended private)
 - IEFMASTERSWA moves master scheduler SWA (CSA-ECSA)
 - IHAACEE3PTY moves selective 3rd party ACEEs (private->extended private)
- Check your DIAGxx VIRTUAL24(x). if anything, understand if it can be removed
- 3.2 Moved nucleus and other modules above the line
- 3.1, 3.2 selective blocks moved above the bar and the journey continues in zOS next
- 2.5 moved allocation and VSAM related blocks above the line

- **3.2 Reduction of abend 40D-10 : Memory termination**

- Not enough private system storage to drive recovery routines resulting in problems
- Automatically, RTM (recovery termination manager) keeps a “buffer” for driving recovery
- Setting DIAGxx CBLOC VIRTUAL31(IHARMPL) can further reduce the chances of out of memory

Less advertised z/OS “resiliency” improvements (Cont)

- 3.2 FRR stack expanded by 15% to prevent out of stack conditions
- 3.2 SETIOS HSWAP,FREEZE : makes it easier to test unplanned PPRC suspension events for Hyperswap
- 3.2 I/O recovery efficiency improvement for RECOVERY PATH_SCOPE=CU
- 3.1, 3.2+ SMF30 virtual and real memory stats improvements for monitoring usage i.e. SMF30_HWMMemlimitMb , SMF30_ObtainsHomeSpace, SMF30_FramesFirstReferenceBacking...
- 3.2 System Logger in memory buffers now reside in Logger High Virtual private memory, with a greatly expanded size limitation of 1 TB
- 3.2 DFSMS: VTOC and INDEX out of space prevention
 - SMS construct can be created to prevent new allocations on volumes below a VTOC/INDEX free threshold
 - LSPACE (macro+SMF19) and IDCAMS DCOLLECT provides VTOC or INDEX usage.

Less advertised z/OS “resiliency” improvements (Cont)

- 3.2 DFSMS: to better manage z/OS storage and avoid disruptive outages, Integrated Catalog and Device Facility (ICKDSF) proactive management enhancement can now create a VVDS at volume initialization rather than when the first catalog or SMS managed dataset is defined.
- zFS shrink function (DASD space reclamation) enhancements: shrink R/O, etc. to satisfy customer ideas.
- JES2: ENQ protecting JES2 checkpoint and spool data sets to prevent inadvertent deletion

Single Engaging place for everything you need to understand and use functions and products!

<https://www.ibm.com/support/z-content-solutions/>

Automation & Management

- ServerPac Installation Using z/OSMF
- IBM z/OS Change Tracker
- Red Hat Ansible Certified Content for IBM Z
- Software update with z/OSMF
- z/OS Management Services Catalog

Modernization

- Automating and shift-left testing for z/OS hybrid applications
- Continuous delivery and deployment
- Continuous integration for the hybrid cloud developer experience
- Discover and plan for z/OS hybrid applications
- EzNoSQL for z/OS
- IBM Z and Cloud Modernization Stack
- IBM Z Distribution for Zowe
- Z Digital Integration Hub
- z/OS Cloud Data Access
- z/OS Connect
- z/OS Container Platform
- z/OS Container Extensions (zCX)
- zCX Foundation for Red Hat OpenShift

Optimization

- Cloud Infrastructure Center
- Cloud Provisioning and Management for z/OS
- Integrated Accelerator for zEDC
- Journey to LinuxONE
- Journey to sustainability with IBM LinuxONE
- Resilience
- System Recovery Boost
- Tailored Fit Pricing for IBM Z

IBM z Content Solutions Center

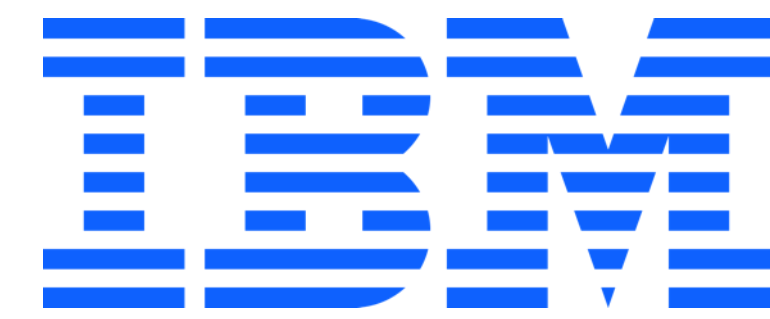
Security

- ServerPac Installation Using z/OSMF
- IBM z/OS Change Tracker
- Red Hat Ansible Certified Content for IBM Z
- Software update with z/OSMF
- z/OS Management Services Catalog

Prediction

- AI Infusion into z/OS
- IBM Open Data Analytics for z/OS
- Journey to AI on IBM Z and LinuxONE
 - Journey to open data analytics
- Machine Learning for IBM z/OS

Currently in IBM z Content Solutions



Thank you

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Citations/Claims/Disclaimers

- 1.Katov PhD, N. (April 2025), Mitigating Fraud in The AI Age: Supporting Transaction Fraud Detection at Scale on IBM z17, Celent
- 2.the-economics-of-technology-investments-downloadTop Ten Payments Companies Processed \$9 Trillion in 2022 Payment Card Volume, GlobeNewswire
- 3.For clients running z/OS v3.1 or higher with a configured high availability IBM software stack on IBM z16 or IBM z17, users can expect up to 99.999999% availability or 315.58 milliseconds of downtime per year when using a GDPS 4.7 Continuous Availability (CA) configuration and workloads. DISCLAIMER: The claim is based on IBM internal data and a GDPS CA three-site configuration, 2 active Sysplex sites and 1 Disaster Recovery (DR) site, consisting of z/OS 3.1 or higher with a Recovery Time objective (RTO) of 2 minutes or less, one of the required GDPS CA IBM middleware stack workloads and replication products running on IBM z16 or IBM z17.
GDPS CA includes resiliency features such as Parallel Sysplex enabled data sharing applications, GDPS Metro Mirror replication (Hyperswap), software replication, and other CA configuration documented high availability features. A supported GDPS CA middleware stack could include CICS v6.2, IMS v15.5, MQ v9.4, and Db2 v13 or at later releases. Clients must follow maintenance, configuration, capacity planning and testing best practices for the entire software stack and hardware configuration. This includes enabling all the resiliency technology for their workloads as defined by GDPS CA, z/OS, and workload related software products. Other configurations may have different availability characteristics.
- 4.500x parameter growth for LLMs [The 7 Biggest Artificial Intelligence \(AI\) Trends In 2022 \(forbes.com\)](#)
- 5.7x Improvement in Compute Throughput [The computing power needed to train AI is now rising seven times faster than ever before | MIT Technology Review](#)
- 6.10x Increase In Energy. [The AI Boom Could Use a Shocking Amount of Electricity - Scientific American](#)
- 7.10x Growth In data due to AI [How AI Developments Impact BI Data Volumes \(kyvosinsights.com\)](#)
- 8.With IBM z17, process up to 450 billion inference operations per day with 1 ms response time using a Credit Card Fraud Detection Deep Learning model .DISCLAIMER: Performance result is extrapolated from IBM® internal tests running on IBM Systems Hardware of machine type 9175. The benchmark was executed with 1 thread performing local inference operations using a LSTM based synthetic Credit Card Fraud Detection model (https://github.com/IBM/ai-on-z-fraud-detection) to exploit the Integrated Accelerator for AI. A batch size of 160 was used. IBM Systems Hardware configuration: 1 LPAR running Red Hat® Enterprise Linux® 9.4 with 6 IFLs (SMT), 128 GB memory. 1 LPAR with 2 CPs, 4 zIIPs and 256 GB memory running IBM z/OS® 3.1 with IBM z/OS Container Extensions (zCX) feature. Results may vary.

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9. The IBM z17 Telum II processor is designed to seamlessly scale peak AI workloads as each core on the chip can access each of the 8 integrated accelerators for AI. By allowing routing of inference requests to any idle IBM Integrated Accelerators for AI within the same drawer, the IBM Integrated Accelerator for AI can increase inference throughput by up to 7.5x as compared to IBM z16.. **DISCLAIMER:** Performance results are based on internal tests exploiting the IBM Integrated Accelerator for AI for inference operations on IBM z16 and z17. On IBM z17, each IBM Integrated Accelerator for AI allows any CPU within a drawer to direct AI inference request to any of the 8 idle AI accelerators on the same drawer. The tests involved running inference operations on 8 parallel threads with batch size of 1. Both IBM z16 and z17 were configured with 2 GCPs, 4 zIIPs with SMT and 256 GB memory on IBM z/OS V3R1 with IBM Z Deep Learning Compiler 4.3.0, using a synthetic credit card fraud detection model (<https://github.com/IBM/ai-on-z-fraud-detection>). Results may vary.
10. Save up to 83% of power consumption by replacing a compared x86 solution comprised of two-year-old servers running AI-infused OLTP workloads with an IBM z17. **DISCLAIMER:** The total cost of ownership (TCO) is based on IBM® internal performance tests running on IBM Systems Hardware of machine type 9175 compared to the same tests running on a commercially available enterprise server with 2x 28 Intel Xeon Gold 5420+ CPU @ 2.20 GHz. The MegaCard benchmark (<https://github.com/IBM/megacard-standalone>) is a containerized IBM WebSphere Liberty v24 application deployed on Red Hat® OpenShift® Container Platform (RHOCP) 4.17 on Red Hat Enterprise Linux® (RHEL) 9.4 with KVM. EDB Postgres for Kubernetes v1.25 is used as the database. The TCO model extrapolated the test results to a typical, complete customer IT solution that includes isolated from each other production and non-production IT environments. TCO included software, hardware, energy, network, data center space, and labor costs. On the IBM z17 side the complete solution requires one IBM z17 Type 9175 MAX 136, and on x86 side, the complete IT solution requires 72 of the compared servers. Results may vary.
11. With IBM z17, process up to 35 billion encrypted requests per day with OLTP applications. **DISCLAIMER:** Performance result is extrapolated from IBM® internal tests running on IBM Systems Hardware of machine type 9175. The Acme Air microservice benchmark (<https://github.com/blueperf/acmeair-mainservice-java>) was deployed on Red Hat® OpenShift® Container Platform (RHOCP) 4.17. The 3 RHOCP Compute nodes ran 3 Acme Air instances in parallel, each driven remotely from Apache JMeter™ 5.2.1 with 128 parallel users. IBM Systems Hardware configuration: 8 LPARs in total with 21 dedicated and 4 shared IFLs (SMT). 3 LPARs running RHOCP Compute nodes each with 7 dedicated IFLs (SMT), 64 GB memory and DASD storage. 5 LPARs each with 4 shared IFLs (SMT), 16 GB memory and DASD storage, providing 3 RHOCP Management nodes and 2 RHOCP Infrastructure nodes. The Network adapters were dedicated for NETH on Linux. Results may vary.
12. Single processor capacity of IBM z17 for equal n-way at common client configurations is approximately 11% greater than on IBM z16 with some variation based on workload and configuration. **DISCLAIMER:** Based on internal measurements. Results may vary by customer based on individual workload, configuration and software levels. Visit LSPR website for more details at: <http://www.ibm.com/support/pages/ibm-z-large-systems-performance-reference>
13. Within each single drawer, IBM z17 provides 20% greater capacity than IBM z16 for standard models and 15% greater capacity on the max config model, enabling efficient scaling of partitions. **DISCLAIMER:** Based on internal measurements. Standard models include IBM z17 Max43, Max90, Max136 and Max183 and IBM z16 Max39, Max82, Max125 and Max168. Max config models are IBM z17 Max208 and IBM z16 Max200. Results may vary by customer based on individual workload, configuration and software levels. Visit LSPR website for more details at: <http://www.ibm.com/support/pages/ibm-z-large-systems-performance-reference>
- 14. The anticipated typical IBM z17 system is estimated to reduce system power consumption by approximately 19% compared to a similarly configured IBM z16 system. **DISCLAIMER:** Based on an expected typical IBM z17 system configuration based on the actual historical average IBM z16 system configuration. IBM z17 is Max90 with 8TB memory, 82 active processors, 5 ICA-SR 2.0, and 3 PCIe+ I/O drawers with 43 I/O adapters. The IBM z16 is configured to provide the same hardware capability. Power consumption is based on the Power Estimation Tool, available at ibm.com/support/resourcelink/api/content/public/PowerEstimationTool-legacy.html. Results may vary.

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